MIL-W-15805F(SHIPS)
10 December 1962
SUPERSEDING
MIL-W-15805E(SHIPS)
20 April 1961

MILITARY SPECIFICATION

WIND DIRECTION AND WIND SPEED INDICATING EQUIPMENT, TYPE B

1. SCOPE

1.1 This specification covers synchro type B wind direction and wind speed indicating equipment for use on Naval Ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- FF-B-171 - Bearings, Ball, Annular; General rpose.
- QQ-A-601 - Aluminum Alloy Sand Castings.

MILITARY
- MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
- MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
- MIL-I-983 - Interior Communication Equipment; Naval Shipboard; Basic Design Requirements For.
- MIL-S-2335 - Synchros, 60 Cycle, 115 volt .
- MIL-M-15071 - Manuals, Equipment and Systems.
- MIL-M-16034 - Meters, Electrical-Indicating Switchboard and Portable Types.
- MIL-E-17555 -.Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of.
- MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification For.

STANDARDS

MILITARY
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
- MIL-STD-242 - Electronic Equipment Parts (Selected Standards).

DRAWINGS

BUREAU OF SHIPS
- 3331238 - Wind Direction and Speed Detector.
- 3331239 - Wind Direction and Speed Transmitter.
- 3331240 - Onboard Repair Parts, Wind Direction and Speed Transmitter, Type B.
- 3331241 - Wind Direction and Speed Indicator.
- 3331265 - Wind Speed Indicator (Cross Wind and Head Wind).
- 3331266 - Crosswind and Headwind Computer Assembly.
- 3331269 - Dials (Wind Speed, Knots (Wind Direction).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed . by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd St., New York 16, New York.)

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. National Electrical Manufacturers Association Specifications

(Application for copies should be made to National electrical Manufacturers Association, 155 East 44th St., New York 17, New York.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards For Federal Services.

(Application for copies shouId readdressed to superintendent of Documents Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Qualification.- The wind direction and wind peed indication equipment furnished under this specification shall be a product which has been tested and passed qualification tests specified herein, and as been listed on or approved for listing on the applicable qualified products list.

3.2 Description. - The wind direction and wind peed indicating equipment shall be designed to give continuous and instantaneous visual indications of wind direction in degrees relative to the ship's bow and relative wind speed in knots. The equipment hall consist of the following:

(a) Wind direction and speed detector.
(b) Wind direction and speed transmitter.
(c) Wind direction and speed indicator or speed recorder(when specified (see 6. 1) .)
(d) Crosswind and headwind computer.
(e) Crosswind and headwind indicator.

3.3 General features. -

3.3.1 The wind direction and wind speed indicating equipment shall be in accordance with the following paragraphs of MIL-I-983, in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern:

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Iron or steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other Metals
Plastics
Ceramics
Impregnating, embedding and encapsulanting compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Gaskets
Dials and pointers
Dial sizes
Locking devices
Washers
Ball bearings
Parts - electrical - general
Use of nonstandard parts
Requirements for semiconductor devices
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Synchros (see 3.4.1, 3.4.2, 3.4.3, 3.4.4.6, and 3.4.5)
Synchro capacitors
Synchro electrical zero and equipment mechanical zero
Servo motors
Electrical tapes
Batteries
Dial illumination lamps
Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuse clips
Metallic rectifiers
Printed wiring and circuits
Enclosures - general
Enclosure - accessibility
Enclosure - degree of
Enclosure - mounting
stiffening grooves
Minimum sheet metal thicknesses (watertight enclosures)
Through bolting
Cable entrance
Cable entrance plates
Ventilation
Size (for submarines and surface ships)

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Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity
Extreme temperature for exposed location
Accelerated life
Salt spray
Shock, vibration and inclination
Features-electrical
Primary power supply circuits
Power supply tolerances
Personnel protect ion
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (Test cable required)
Terminal boards, connectors and terminals
Synchro connections and markings
Wiring
Color coding
Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise (grade C)
Structureborne noise
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Drawings - manufacturing
Bill of materials
Interchangeability and standardization
Manuals (see 3.3. 1.2)
Repair parts
Repair parts (electronic)
Designation and marking (serial numbers required)
Reports
Item names and nomenclature
Workmanship and general examination

3.3. 1.1 Electron tubes. - Electron tubes shall be selected from the preferred types listed in MIL-STD-242.

3.3.1.2 Manuals - Manuals shall be in accordance with type I of MIL-M-15071. Deviations from previously approved existing manuals, except for the approval procurement and record page, shall not be made without specific authorization by the Bureau of Ships.

3.3.2 Accuracy and response. -

3.3.2.1 The equipment shall be designed so that the error of the wind speed indicator readings shall not exceed the following values:

(a) 3 to 45 knots plus or minus 1-1/2 knots.
(b) 45 to 60 knots plus or minus 2 knots.
(c) 60 to 80 knots plus or minus 2-1/2 knots.
(d) 80 to 100 knots plus or minus 3 knots.
(e) Maximum starting wind speed -3 knots.
(f) Maximum stopping wind speed -2 knots.

3.3.2.2 The time response of the wind speed transmitter shall be within the limits shown on figure 1.

3.3.2.3 The equipment shall be designed so that the error of the wind direction indicator readings shall not exceed the following values:

(a) 1 to 3 knots minus 5 degrees.
(b) 3 to 8 knots minus 4 degrees.
(c) 8 to 100 knots plus or minus 2 degrees.

3.3.2.4 The response of the wind direction indication shall be, for any change in wind direction, at the rate of 6-1/2 degrees ± 1-1/2 degree per second.

3.3.3 Continuous operation. -

3.3.3.1 All wind speed components shall be capable of continuous operation at full scale reading.

3.3.3.2 All wind direction components shall be capable of continuous rotation at a speed of 5 revolu - tions per minute (r.p.m.).

3.4 Detail requirements. -

3.4.1 Wind direction and speed detector .- The detector shall be manufactured in accordance with Drawing 3331238. The detector shall provide a means for generating and transmitting electrical signals, indicative of wind direction and wind speed, to the associated transmitter. The detector shall consist of a streamlined vane and a three-blade impeller mounted on a vane support (see figure 2). Any movement of the vane and impeller blades shall be transmitted mechanically to size 1 HG synchro generator in accordance with MIL-S-2335, housed in the vane or vane support. The design shall be such that 100 knots of wind speed shall equal 190 r. p. m. of the speed synchro generator. The wind vane and vane support shall be of splashproof

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obstruction fabricated of nickel-copper alIoy. The exposed nickel-copper alloy shall not be painted, but shall be finished by a sandblast using silica powder-825 mesh with an air line pressure of approximately 80 pounds per square inch (p.s. i.) and a nozzle size of 1/2- to 3/4-inch diameter. The impeller blades shall be of molded plastic of type CFG of MIL-M-14. The impeller shall contain a nickel-copper alloy nose cap with captive screw. A connector assembly arrangement containing a Pyle National connector, types AP-810 and AR-810, or equal shall be provided to permit removal of the detector mechanism without removing the base from the mounting pintel or disturbing the ship's wiring cable. A raincap shall be provided to protect the connector when the detector mechanism is removed. The connector housing shall contain a 3/4-inch National Pipe thread, class 2 in accordance with Handbook H-28, pipe-plug for optional cable connector. The design of the detector shall be such as to permit easy disassembling for boiling, cleaning, inspection, or replacement.

3.4.1.1 Mounting. - The detector shall be designed for mounting on the end of a l-inch diameter vertical pintel. The mounting arrangement shall provide a clamping fit and one 5/16-inch through bolt (see figure 2).

3.4.1.2 Balance - The wind vane shall be statically balanced to within 5 ounce-inches of true static balance. The impeller shall be statically balanced to within 10 gram-inches.

3.4. 1.3 Rotor starting torque. - With the detector mounted in the normal manner, the rotor shaft shall start to turn at a torque of 0.056 inch-ounce.

3.4.1.4 Direction vane torque. - With the detector mounted in the normal manner with its support horizontal, the vane shall start to rotate at a torque of 4.0 inch-ounces.

3.4.1.5 Size. - The complete detector assembly shall weigh not more than 21 pounds, and shaIl not exceed the dimensions shown on figure 2.

3.4.1.6 Shock - The detector assembly will not be subjected to the shock requirements as outlined in MIL-I-983.

3.4.1.7 Bearings - All bearings, including those used in the synchros, shall be "sealed-for-life" type, FF-B-171, annular ball bearings.

3.4. 1.8 Electrical contact, alloy. - The composition of the stationary electrical contacts shall be a fine silver graphite alloy, 97 percent silver, 3 percent graphite. The material of the rotating contact shall be a silver alloy, 90 percent silver, 10 percent copper. Contact pressure shall be 20 ± 2 grams when tested with a spring scale pull. Mercury shall not be used in the fabrication of electrical contact parts, terminals, or terminal Iugs.

3.4.2 Wind direction and speed transmitter - The wind transmitter shall be manufactured in accordance with Drawing 3331239. The wind direction and speed transmitter shall contain, in a dripproof enclosure designed for bulkhead mounting, the mechanisms necessary to convert the input from the wind direction and speed detector to relative wind direct ion in degrees and relative wind speed in knots, and to transmit this intelligence to wind direction and speed indications. The transmitter shall be designed so that it shall be possible to remove and repair either the wind direction or wind speed assembly of the mechanism without disturbing the operation of the other assembly. The speed unit shall be a roller disc type integrator which measures the rotary motion of a receiving type 1 F synchro, in accordance with MIL-S-2335, and displaces a transmitting' type 5G synchro, in accordance with MIL-S-2335, at a rate (see figure 1) angularly proportional to rotor speed of the detector, that is, wind speed. The discs shall be powered by a synchronous motor, Holtzer-Cabot model RBC 2505, or equal, which shall receive its power from the ship's service 120-volt, 60-cycle, a a.c. supply. A low-limit cut-out switch shall be provided for breaking the circuit to the constant speed synchronous motor when the roller is at the center of the disc, less than l/2-knot speed. The direction unit shall contain a static type amplifier employing either magnetic amplifier or solid state techniques which receives angular displacements of the vane of the detector. These angular displacements shall be retransmitted to the indicator by a type 5G synchro, in accordance with MIL-S-2335, at a reduced rate of 6-1/2 ± 1.5 degrees per second. As retransmitted angular displacements are approximately constant and are not at the same rate as the angular displacements of the vane, damped indications are provided. Means shall be incorporated in the wind direction assembly for adjusting the electrical position of the aloft detector with respect to the fore and aft axis of the ship. The units shaIl be bolted in the case, and terminal boards shall be provided in the case for incoming wires. no holes fitted with 1-inch National pipe thread class 2 in accordance with Handbook H-28 pipe plugs shall be provided for attaching stuffing tube fittings. Electrical connections shall be made to each unit by plug-jack type terminals, Winchester types A18P and A18S, or equal. Power supply tolerances as specified in MIL-I-983 shall apply. In order to provide a means for adjusting the transmitter, a method or methods shall

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be incorporated in the wind direction and wind speed mechanisms to indicate accurately the magnitude of the transmittcd wind direction and wind speed signals.

3.4.2.1 Sensitivity - The wind direction assembly shall be sufficiently sensitive so that when the wind vane is displaced 8 degrees from a 5-knot wind, it shall, upon release, meet the accuracy requirements of 3.3.2.3.

3.4.2.2 Size. - The transmitter shall be as small and Iightweight as practicable. but in no case shall dimensions and weight exceed the following exclusive of terminal tubes:

Height - 13-1/ 2 inches.
Width - 15-1/ 2 inches.
Depth - 7-1/2 inches.
Weight - 42 pounds.

(Figure 3 is provided for guidance only)

3.4.3 Wind direction and speed indicators. - The wind indicator shall be manufactured in accordance with type B of Drawing 3331241. The indicators shall contain, in a watertight enclosure, the mechanism incorporating size 1 F synchros, as specified in MIL-S-2335. to receive signals from the wind direction and speed transmitter, and to indicate the relative wind direction in degrees and the apparent wind speed in knots. The indicator shall contain two independent units. one for direction and the other for speed. Electrical connections shall be made to each unit by plug-jack type terminals, Winchester types A7P and A7S, or equal. The units shall be bolted in the case. A hole in the case and plug with a l-inch National pipe thread class 2 in accordance with Handbook H-28 shall be provided in one end for a stuffing tube fitting.

3.4.3.1 Dials .- The indicator shall have two dials 4 inches in diameter; one for indicating wind direction and the other for indicating wind speed. The dials shall be marked in accordance with Drawing 3331269. Red dial illumination shall be provided in the indicator.

3.4.3.2 Size. - Indicators shall be as small and lightweight as practicable, but in no case shall dimensions and weight exceed the following:

Height - 13-3/4 inches.
Width - 7-1/2 inches.
Depth - 6 inches.
Weight -23 pounds.

(Figure 4 is provided for guidance only)

3.4.3.3 The transmitters and Indicators provided under this spec.fication shall be equipped with internal terminal boards to facilitate the ship's wiring.

3.4.3.4 Fasteners - Fasteners manufactured from non -ferrous and corrosion-resisting steel alloys shall be used. No carbon or alloy steel fasteners in any form shall be used.

3.4.3.5 Enclosure - The enclosure shall be fabricated of cast aluminum alloy conforming to Commercial designation 356 T-6 or class 3M, type T-6 of QQ-A-601. One coat of zinc chromate primer followed by two coats of high grade semi-gloss light gray equipment enamel. The anodizing, primer and enamel coats shall be applied in accordance with the best commercial practice.

3.4.3.6 Illumination. - The illumination transformers for the indicator lamps shall conform to the applicable National Electrical Manufacturers Association specification requirements. A power transformer for the Iighting circuit shall be located in the housing. A variable resistor for controlling the light intensity shall be provided, and shall be adjusted by a knob from the outside. The housing shall be provided with a cover which has two clear ports over the dials.

3.4.4 Wind direction and wind speed recorder. -

3.4.4.1 The wind direction and wind speed recorder shall be a two-element recorder that simultaneously produces, in separate channels, inked traces of wind direction and speed values on a continuous paper chart, when connected electrically to a wind direction and speed transmitter. The recorder mechanism basically shall consist of a direction section, a speed section, an electrical zeroing circuit, a chart drive mechanism, a chart guide - pen lift mechanism, and a chart.

3.4.4.2 Case. - The recorder mechanism shall be bound in a cast-aluminum case suitable for shelf or three point bulkhead mounting. The outside dimensions shall be approximately 16 inches wide by 14 inches high by 11 inches deep. A gasketed door on the front shall give access to all assemblies and parts. The door shall have a transparent window allowing full view of the recorder chart, pen tips, and scales. The case shall have internal illumination controlled with a switch. One electrical connector shall be mounted in a well in the lower rear part of the case. (Figure 5 is provided for guidance only.)

3.4.4.3 Connections and wiring. - The connection and wiring shall be in accordance with figure 6. The power and signal circuits shall be energized through a connector, Jones Catalogue No. P2410-SB, or equal. A mating connector, Jones Catalogue No. S2410-OCT, or equal, shall be provided for ship's wiring.

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3.4.4.4 Wind direction mechanism. - The wind direction mechanism shall be a subassembly consisting of a synchro receiver driving the pen, and miscellaneous parts and adjustments as required.

3.4.4.5 Accuracy. - The static error at all positions of the pen shall not exceed plus or minus 4 degrees.

3.4.4.6 Synchro - The synchro shall be a torque transmitter conforming to type 1 F of MIL-S-2335. The synchro shall be mounted by its mounting flange with a positive clamping arrangement allowing adjustment of not less than 360 degrees. The synchro mount shall allow the synchro to be replaced without loosening or removing the wind direction mechanism from the case. The synchro shall be coupled to the open drive through a gear reduction. The synchro hall have a zero adjustment with a range of 26.5 plus or minus 5 degrees.

3.4.4.7 Chart scale. - The scale for the wind direction mechanism shall have an associated scale calibrated 0 to 360 to 180 degrees in 10-degree incrrements. Major calibration marks shall be at multiples of 30 degrees, and 90-degree intervals,

3.4.4.8 Wind speed mechanism. - The wind peed mechanism shall be a subassembly consisting of synchro receiver driving the pen, a pen repositioner, and miscellaneous parts and adjustments as required.

3.4.4.9 Accuracy. - The error of the wind speed mechanism shall not exceed that shown in 3.3.2.

3.4.4.10 Response - The response of the wind peed recorder shall be within the limits shown in .3.2.

3.4.4.11 Chart scale. - The chart scale for the wind speed mechanism shall have an associated scale calibrated from 0 to 100 knots in 2-knot increments. Major calibration marks shall be at multiples of 10 nets, and identifying numbers at intervals of 20 nets.

3.4.4.12 Chart drive mechanism. - The chart rive mechanism shall be a subassembly consisting f a chart supply bin, a chart drive roll driven by a synchronous motor, a reroll spool driven by a stalled reduction motor, and mechanical parts and electrical arts as required. Not less than 6 inches of the latest recorded chart shall be visible. The chart rive mechanism shall accept ships standard 115-volt, 60-cycle power.

3.4.4.13 Chart speeds. - The chart drive mechanism shall advance the chart at 3 inches per hour when energized. Readily replaceable gears shall provide for a chart speed of 6 inches per hour.

3.4.4.14 Drive roll.- The drive roll shall have three sets of chart drive pens around the roll with center to center spacing of the sets 5.125 inches ± 0.010 inch. The drive roll shall be driven through a friction slip clutch, and shall have an adjusting wheel on the right.

3.4.4.15 Chart drive motor. - The chart drive motor shall be a synchronous motor operated at 1 revolution per minute when energized from a 60-cycle power source. Its torque shall be not less than 30 inch-ounces, when operated from 120 volts.

3.4.4.16 Reroll spool. - The reroll spool shall have a slot for starting the roll. The rerolled chart shall be readily removable from the chart drive mechanism.

3.4.4.17 Reroll motor. - A reroll motor shall be provided. The reroll motor shall be de-energized automatically when the reroll spool is removed.

3.4.4.18 Chart - The chart shall be 10 3/4 inches wide by 100 feet long. The chart shall be printed on white paper and shall have two sets of red or orange calibrations, wind direction on the left, and wind speed on the right. The calibrations shall match the scales, and shall be numbered at intervaIs of 6 inches. The direction calibration shall be 4.513 inches wide, and the speed calibration shall be 4.500 inches wide. The two sets of calibrations shall be separated by 0.375 inch ± 0.010 inch. There shall be three rows of drive holes, a row of round holes on the centerline of the chart, and a row of holes, elongated in the width direction, near each edge. The distance between centerlines **of** the holes shall be 5.15 inches and 0.5 inch.

3.4.4.19 Time calibration. - The chart calibration shall have red arc lines corresponding to the pen arcs at 0.5-inch intervals on the length of the chart. Every sixth arc shall be heavier, and shall be identified with time in l-hour increments in the 12-hour system. The chart shall have the remaining lengths marked in feet at l-foot intervals.

3.4.5 Crosswind-headwind computer. - The crosswind computer shall be manufactured in ac - cordance with Drawing 3331266. The crosswind computer shall be contained in a dripproof enclosure designed for bulkhead mounting the mechanisms necessary to convert the output signals from the wind direction and speed transmitter into two d.c. voltages proportioned to the headwind and crosswind components of the relative wind for both straight

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and angle deck. These outputs shall be capable of driving a total of five indicators. The signals from the wind direction and speed transmitter shall be received by type lHCT synchros, in accordance with MIL-S-2335. The servo amplifier shall be a solid state device.

3.4.5.1 Size - The crosswind-headwind computer shall be as small and lightweight as practical, but in no case shall the dimensions and weight exceed the following:

Height - 13 1/2 inches
Width - 15 1/2 inches
Depth - 7 1/2 inches
Weight -35 pounds
(Figure 7 is provided for guidance only)

3.4.5.2 Accuracy - The crosswind-headwind computer shall resolve the input signal so that the accuracy of the output signal voltage shall be within ± 1 knot of the theoretical computed value.

3.4.5.3 Sensitivity .- When the input signal is changed instantaneously from 0 to 60 knots the output signal shall reach this value in a time not to exceed 12 seconds.

3.4.6 Crosswind-headwind indicator. - The crosswind-headwind indicator shall be manufactured in accordance with Drawing 3331265. The crosswind-indicator shall be contained in a watertight enclosure designed for bulkhead mounting two voltmeters made in accordance with MIL-M-16034, and which shall have an accuracy of ± 1 percent. The upper voltmeter shall be calibrated 30 to 0 to 30 knots crosswind, and the lower voltmeter shall be calibrated 0 to 60 knots headwind. Red dial illumination with a rheostat to vary the intensity shall be provided. Two pilot lights shall serve to indicate whether the dials read straight or angle deck values.

3.4.6.1 Size. - The crosswind-headwind indicator shall be as small and lightweight as practicable, but in no case shall the dimensions and weight exceed the following:

Height - 13 3/4 inches
Width - 7 1/2 inches
Depth -6 1/2 inches
Weight -24 pounds
(Figure 8 is provided for guidance only).

3.4.7 Repair parts. - Repair parts and applicable repair parts lists required for maintenance of the parent equipment shall be in accordance with Drawing 3331240. The items and quantity of each, required for onboard repair parts shall be as shown in table 1.

Table I - Onboard repair parts required per ship.

Item	Part	MOD 0	MOD 1	MOD 2
	Detector, transmitter indicator			
1	Ball bearings (one of each type used in the detector) (only if non-standard)	1	1	1
2	Pliers "tru-arc" (type 62562-A)	1	1	1
3	Dampener assembly (transmitter)	0	1	0
4	Integrator assembly (transmitter)	0	1	0
5	Box special screws (screws not available in normal stock)	1	1	1
	Crosswind and headwind computer			
6	Potentiometer (one of each)	0	1	0
7	Motor assembly (Crosswind and headwind indicator)	0	1	0
8	Microammeter movement	0	1	0
9	Roller disc	1	1	1

3.4.8 Drawings .- Drawings shall be furnished in accordance with MIL-D-963 class A, shop or manufacturing type drawings are required. 'The following information shall supplement the drawing information requirements currently specified under MIL-D-963:

(a) Power requirements under full load and standby conditions.
(b) Heat dissipated under full load and standby conditions.
(c) Voltage and resistance readings at key service points.
(d) All parts in correct relation as to physical location with terminals clearly shown with markings and all wiring between parts.
(e) The center of gravity shall be shown for each piece of parent equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless Otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of.

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the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests 1/. Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. One complete set of equipments consisting of one each speed detector, speed transmitter, either a speed indicator or speed indicator recorder, and crosswind-headwind computer and indicator shall be operated and tested as a wind indicating system, and shall be subjected to the following qualification tests. Tests shall be performed in general in the order listed:

Test	Requirement paragraph	Test paragraph
Operating		
All except crosswind and head wind equipment	3.3.1	4.5.1
Crosswind and headwind equipment	3.4.5	4.5.1.5
Accuracy (first part)		
All except crosswind and headwind equipment	3.3.2	4.5.2.1
Crosswind and headwind equipment	3.4.5.2	4.5.1.5
Supply line voltage and frequency variation	3.3.1	MIL-I-983
Shielding and radio frequency noise reduction	3.3.1	MIL-I-983
Airborne noise	3.3.1	MIL-I-983
Temperature and humidity	3 . 3 . 1	MIL-I-983
Inclination	3.3.1	MIL-I-983
Accelerated life 2/	3.3.1	MIL-I-983 and 4.5.3
Accuracy (second part)	3.3.2	4.5.2.2
Angular displacement retransmission	3.4.2	4.5.4
Sensitivity		
All except crosswind and headwind equipment	3.4.2.1	4.5.4
Crosswind and headwind equipment	3.4.5.3	4.5.1.5
Dielectric strength	3.3.1	MIL-I-983
Insulation resistance	3.3.1	MIL-I-983
Size	3.4.2 .2; 3.4.3.2; 3.4.5 .1; 3.4.6.1 and figure 2	
Mounting	3.4.1.1	MIL-I-983 and 4.5.5
Saltspray (not required for transmitter or. crosswind and headwind computer)	3.3.1	MIL-I-983
Vibration (vital)	3.3.1	MIL-I-983
Shock (not required for recorder)	3.3.1	MIL-I-983
Accuracy (third part)	3.3.2; 3.4.2.1; and 3.4.5.2	4.5.2.3 and 4.5.4

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

2/ A start stop condition that will duplicate actual in-service operation shall be introduced for this phase of the test. A minimum of twenty "starts-stops" at the end of each hour's test is required. The deenergized period shall be of sufficient duration so as to permit the equipment under test to come to a complete rest. The rest period shall not exceed 30 seconds.

4.3 Sampling for quality conformance inspection. -

4.3.1 Inspection lot. - All equipment offered for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Group A tests. - All equipment furnished on the contract or order shall be subjected to the group A test listed in 4.4

4.3.3 Sampling for group B test .- A sample of equipments shall be selected from each lot in

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accordance with table II and shall be subjected to the group B test specified in 4.4 with quality conformance based on the requirements of MIL-STD-105.

Table II - Sampling for group B tests. .

Number of equipments in Inspection lot	Number of equipments in sample	[Number of equipments nonconforming on any group B test)	
		Acceptance number	Reject ion number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.3.4 Sampling for group C tests. - Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests listed in 4.4.

4.3.5 Group A and group B tests. - The results of each test shall be completed with the requirements of this specification. Failure to conform to the requirements for any test shall be counted as a defect and the equipment shall not be offered for delivery. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for the rejection of the lot represented by the sample.

4.3.6 Group C hests.- The results of each test shall be compared with the requirements of this specification. In the event of failure to conform to the requirements of this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency on all equipments produced on the contract or order.

4.4 Quality conformance inspection - The samples specified in 4.3.2, 4.3.3, and 4.3.4 shall be subjected to the following group A, group B, and group C tests, as applicable. Tests shall be performed in general in the order listed.

Test	Requirement paragraph	Test paragraph
Group A		
Operating		
All except crosswind and headwind equipment	3.3.1	4.5.1
Crosswind and headwind equipment	3.4.5	4.5.1.5
Group B		
Supply line voltage and frequency variation	3.3.1	MIL-I-983
Group C		
Airborne noise	3.3.1	MIL-I-983
Accuracy (first part)		
All except crosswind and headwind equipment	3.3.2	4.5.2.1
Crosswind and headwind equipment	3.4.5.2	4.5.1.5
Radio frequency noise interference	3.3.1	MIL-I-983
Temperature and humidity	3.3.1	MIL-I-983
Inclination	3.3.1	MIL-I-983
Accelerated life 1/	3.3.1	MIL-I-983 and 4.5.3
Angular displacement retransmission	3.4.2	4.5.4
Sensitivity		
All except crosswind and headwind equipment	3.4.2.1	4.5.4
Crosswind and headwind equipment	3.4.5.3	4.5.1.5
Accuracy (second part)	3.3.2	4.5.2.2

See footnote on next page.

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Test	Requirement paragraph	Test paragraph
Group C (cont'd)		
Size	3.4.2.2; 3.4.5.1; 3.4.6.l and figure 2	
Mounting	3.4.1.1	4.5.5 and MIL-I-983
Salt spray (not required for transmitter and crosswind and headwind equipment)	3.3.1	MIL-I-983
Vibration (vital)	3.3.1	MIL-I-983
Shock (not required for recorder)	3.3.1	MIL-I-983
Accuracy (third part)	3.3.2; 3.4.2.1; and 3.4.5.2	4.5.2.3 and 4.5.4

1/ A start .and stop condition that will duplicate actual in-service operation shall be introduced for this phase of the test. A minimum of twenty "start-stops" at the end of each hour's test is required. The deenergized period shall be of sufficient duration so as to permit the equipment under test to come to a complete rest. The rest period shall not exceed 30 seconds.

4.5 Test procedures. -

4.5.1 Operating test. - The operating test shall resist of the following for the individual components:

4.5. 1.1 Final inspection procedure for wind direction and speed transmitter (detector). - Final inspection procedure for wind direction and speed transmitter (detector) shall be as follows:

(a) General examination in accordance with MIL-I-983 and as follows: Visual examination of unit, covering finish, identification plate, installation of lockwashers, and similar items.
(b) Dielectric strength in accordance with MIL-I-983 and as follows: Test with Hi-Pot (set at 900 volts) with one lead connected to support assembly, and other lead connected successively to each prong of connector for 5 seconds.
(c) Insulation resistance in accordance with MIL-I-983 and as follows: Connect one lead of rnegger to support assembly and other lead to each prong of connector. The resistance shall be not less than 10 megohms at 500 v.d.c.
(d) Direction of rotation and electrical zero position shall be conducted as follows:
    (1) Install transmitter on test standard and connect to test indicator. Position vane to North position and check test indicator to within plus or minus 1-1/2 degrees. Move to each cardinal point and check direction of rotation and accuracy. Frictional torque 4.0 inch-ounce (direction) and 0.056 inch-ounce (speed) shall be verified.
    (2) Rotate speed shaft counterclockwise by hand, and check direction of indicator.
    (3) Adjust each brush to bear on contact slip ring with a force of 15 to 20 grams.

4.5. 1.2 Final inspection procedure for wind direction and speed master transmitter (transmitter). - Final inspect ion procedure for wind direction and speed master transmitter (transmitter) shall be as follows:

(a) Assembled case with terminal strips, receptacles and wiring.
    (1) General examination in accordance with MIL-I-983 as follows: Visual examination of case assembly covering finish, installation of Iockwashers, soldered connections, and similar items.
    (2) Continuity: With Simpson meter (scale (RX1) check for continuity between Winchester receptacles and terminal strips.
    (3) Dielectric strength in accordance with MIL-I-983.
        a. Position Hi-Pot test plug to a Winchester receptacle. Set Hi-Pot to 900 volts and connect one lead to test plug and other lead to spacing collar of Winchester receptacle.
        b. Test for 5 seconds and then repeat on other connector.
    (4) Insulation resistance in accordance with MIL-I-983.

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4.5.1.2 (continued)

a. Connect one lead of the megger to the spacing color of the Winchester receptacle. Connect the other lead to each terminal of the two terminal strips successively. The resistance shall be not less than 10 megohms at 500 v.a.c.

(b) Speed transmitter:

(1) General examination in accordance with MIL-I-983 and as follows: Visual examination speed unit covering finish, installation of lockwashers, soldered connections.

(2) Dielectric strength in accordance with MIL-I-983 and as follows: Position Hi-Pot test plug to Winchester plug. Set Hi-Pot to 900 v. and connect one lead to test plug, and the other lead to supporting post . Test for 5 seconds.

(3) Insulation resistance in accordance with MIL-I-983.

a. Connect one lead of megger to supporting post of unit. Connect the other lead successively to each prong of Winchester plug. The resistance shall be not less than 10 megohms at 500 v.d.c.

(4) Operational test:

a. Connect unit to test set. Energize from line.

b. Switch test set to electrical zero. The 5G synchro should read zero.

c. Switch to speed test and check the 5G synchro readings as specified in table III.

NOTE: The values shown under the "correct intensity knots" apply for nominal voltage (115 volts) and frequent y (60 cycles). At frequencies other than 60-cycle, the values shall be proportional to the change in frequency. All values shall be recorded, for example:
60 cycles/1800 r.p.m., 76 knots.
57 cycles/1800 r.p.s.,
$\frac{57}{60}$ of 76) 80 knots.

Table III - Synchro readings.

Test set speed (r.p.m.)	Correct intensity knots	Permissible error plus or minus
400	17.5	3/4
800	34.5	3/4
1200	51.0	1
1600	67.5	1-1/4
1800	76.0	1-1/4

d. Test set to consist of a 1F synchro driven by variable speed mechanism with switch for electrical zero position.

(c) Direction transmitter in accordance with MIL-I-983:

(1) General examination:

a. Same as (b) (l).

(2) Same as (b) (2).

(3) Same as (b) (3).

(4) Operational test:

a. Connect unit to test set. Energize from line.

b. Switch test set to electrical zero. The 5G synchro should read zero.

c. Switch to direction test and check direction of rotation and accuracy at cardinal points of plus or minus 1-1/2 degrees.

d. Test set to consist of a lF synchro with dial and pointer with switch for electrical zero position.

(d) Complete assembly:

(1) General examination in accordance with MIL-I-983:

a. Visual examination of finish, identification plate, screw, and lockwashers.

4.5.1.3 Final inspection procedure for wind direction and intensity indicator. - Final inspection procedure for wind direction and intensity indicator shall be as follows:

(a) Assembled case, with terminal strips, receptables and wiring, rheostat, and transformer:

(1) General examination in accordance with MIL-I-983:

a. Visual.

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4.5.1.3 (continued)

b. Continuity check of wiring.
(2) Dielectric strength in accordance with MIL-I-983: Hi-Pot at 900 volts for 5 seconds. 1/
(3) insulation resistance in accordance with MIL-I-983: Not less than 10 megohms at 500 v.d.c.
(4) Operation of rheostat.

(b) Assembled and wired unit (direction or speed):
(1) General examination in accordance with MIL-I-983:
a. Visual.
b. Continuity check of wiring.
(2} Dielectric strength in accordance with MIL-I-983: Hi-Pot at 900 volts for 5 seconds.
(3) Insulation resistance in accordance with MIL-I-983: Not less than 10 megohms at 500 v.d.c.
(4) Accuracy:
a. Check electrical zero - plus or minus 0.5 degree.
b. Check direction of rotation.
c. Check lights.

(c) Assembled case and units.
(1) Visual check:
a. Finish, screws.
b. Identification plate.

4.5.1.4 Final inspection procedure for wind direction and speed recorder:

(a) General examination in accordance with MIL-I-983.

4.5.1.5 Final inspection procedure for cross-wind computer and indicator.

(a) General examination in accordance with MIL-I-983.
(b) Continuity check of wiring.
(c) Insulation resistance test (MIL-I-983).
(d) Operational test as foIlows:

Connect crosswind computer to crosswind indicator. ‘With a test set to furnish synchro signal inputs equivalent to the signals set out by the transmitter, check he indicator for minimum and maximum crosswind ad headwind for both straight deck and angel deck. Accuracies shall be checked for conformance with 3.4.5.2. Response time shall be checked for conformance with 3.4.5.3.

1/All synchros, transistors, and similar parts that may be damaged by excessive voltage shall be disconnected during this test.

4.5.2 Accuracy - Accuracy tests shall be conducted in three parts as follows:

4.5.2.1 Accuracy - first part. - An initial accuracy test on the complcte wind system shall be conducted in a wind tunnel at the National Bureau of Standards (under auspices of the Naval actlvlty conducting the tests) to determine if the equipment complies with the accuracy requirements specified in 3.3.2

4.5.2.1.1 The complete wind speed system shall be operated at starting and stopping speeds and each wind speed corresponding to 10-knot increments of wind speed over its entire range, and shall not show an error in excess of that specified in 3.3.2.1.

4.5.2.1.2 The position of the wind vane shall be varied by 30-degree increments from O degree to 360 degrees, and for each position of the vane. The wind direction indicators shall meet the requirements specified in 3.3.2.3.

4.5.2.1.3. From the data obtained in tests specified in 4.5.2.l.l and 4.5.2.1.2, curves of error versus speed and direction shall be plotted to rectangular coordinates and incorporated in the laboratory report. Also, a calibration curve or r.p.m. of the impeller versus wind speed shall be included in the report.

4.5.2.2 Accuracy - second part. - Upon completion of the first 50 hours of the accelerated life test, an accuracy check shall be made to determine if the equipment, without adjustment, meets all the accuracy requirements specified in 3.3.2.

4.5.2.3 Accuracy - third part. - Upon successful completion of all tests, an accuracy test on the complete wind system shall be conducted in a wind tunnel at the National Bureau of Standards (under auspices of laboratory personneI) to determine that the equipment complies with the accuracy, angular displacement retransmission, and sensitivity y requirements specified in 3.3.2 and 3.4.2.1, respectively .

4.5.3 Accelerated life. - During the accelerated life test specified in MIL-I-983, the equipment shall be operated at constantly varying impeller speeds, oscillating through the range of 10 to 95 knots at the rate of change of speed of approximately 15 knots per minute, except that once each hour the impeller speed shall be steadied at approximately 95 knots for a sufficient length of time to permit determination of indicator error. At the same time, the wind vane support shall be oscillated through 120 degrees at the rate of 1/2 cycle per minute to provide a repeated reversal of the follow-up motor in the

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transmitter. Upon completion of the 500-hour test, the detector shall be disconnected from the system and, with the wind vane locked in place, the vane support shall be rotated at a rate of 60 r.p.m. for a period of 4 hours. At the end of the 4-hour period, the procedure shall be repeated in the reverse direction for a period of 4 hours.

4.5.4 Angular displacement retransmission and sensitivity test. - Upon completion of the first 50 hours of the accelerated life test, damping and sensitivity tests shall be conducted as folIows:

4.5.4.1 With the detector operating in a normal position, set the wind vane to indicate a definite direction and de-energize the system. While de-energized rotate the wind vane 120 degrees, and with the vane in this position energize the system, nothing the time required for the wind direction indicator to become synchronized. The time required (7.5 degrees plus or minus 1.0 degree per second) shall meet the requirements specified in figure i .

4.5.4.2 With the detector in a position so that the plane surface of the wind vane is horizontal, verification of the requirement of 3.4.1.3 shall be made.

4.5.4.3 Maximum frictional torque of detector rotor shaft shall conform to 3.4.1.3.

4.5.5 Mounting - Compliance with the requirements of 3.4.l.l through 3.4.1.5, as well as special features of the various component mountings shall be determined in general accordance with MIL-1-983.

4.5.6 During the entire accelerated life, shock, and vibration tests, no indicator shall show an error in excess of that specified in 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts. -

5.1.1 Wind direction and wind speed indicating equipment. -

5. 1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing - Packing shall be accomplished in a manner—which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall Include nomenclature, contract or order number, contractor's name, and destination.

5.1.2 Onboard repair parts. - The onboard repair parts shall be preserved and packaged level A; packed level C, and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, and packing shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection durng domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity.

5.2.1.1 Preservation, packaging, packing, and marking. - The equipment and accessories, repair parts and technical publications shall be preserved and packaged level A or C; packed level A or B as specified and marked in accordance with MIL-E-17555).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity and type of items required. (See 3.2.)
(c) Quantity of manuals required.
(d) Whether certification data sheets are required.

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(e) Whether unit SNEN is to be entered on identificat ion plate.
(f) Inventory control point for repair parts.
(g) Whether MOD 0, MOD 1, or MOD 2 repair parts are required (see 3.4.7).
Level of preservation, packaging, and packing, and marking if other than as specified in 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-15805 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.

Notice .- When Government drawings, specifications. or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6660-N008Sh)

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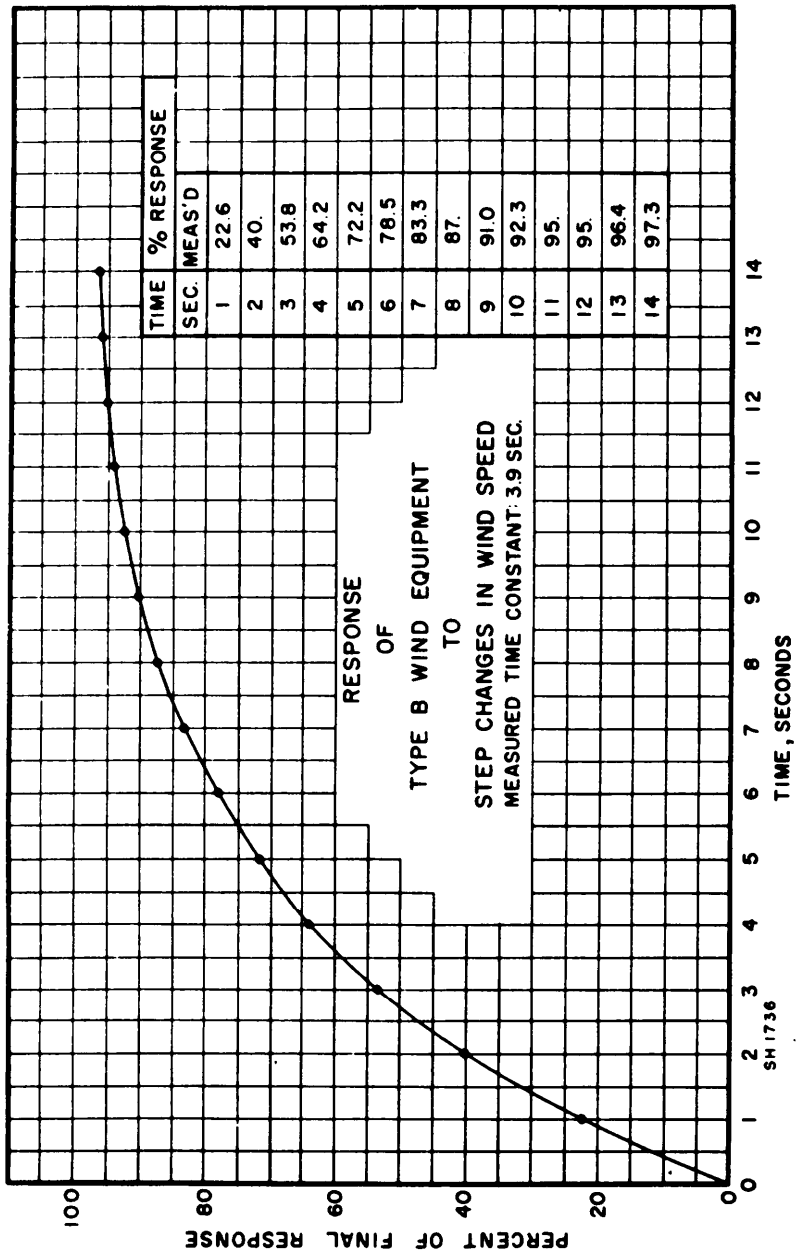


Figure 1 - Response of wind equipment.

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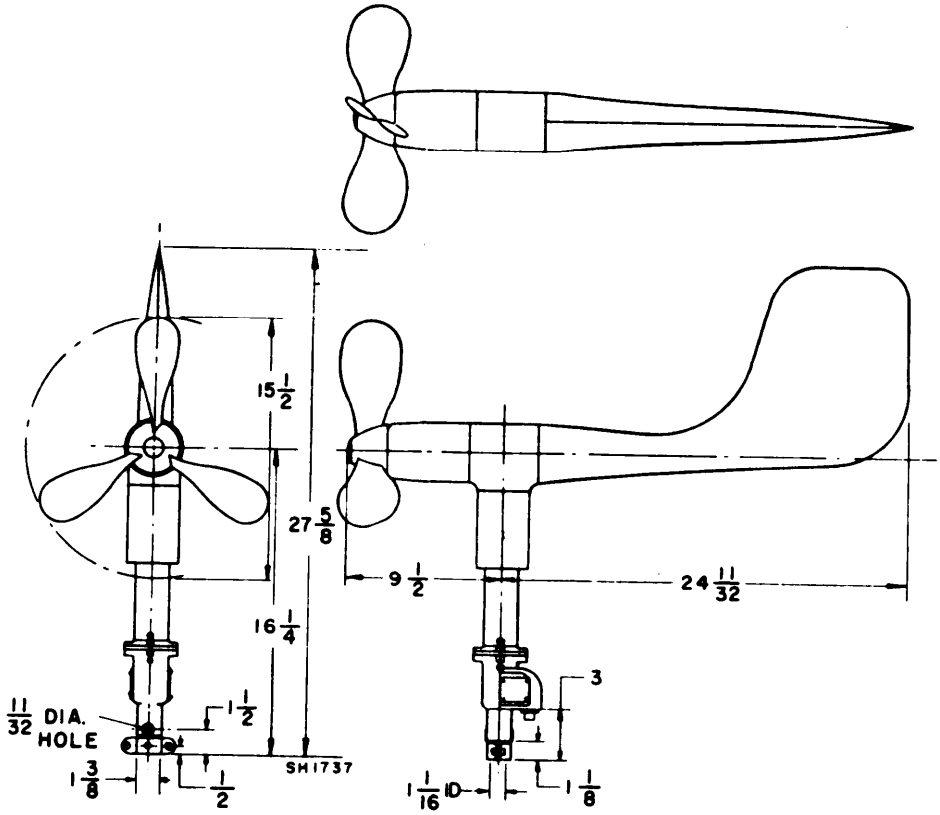
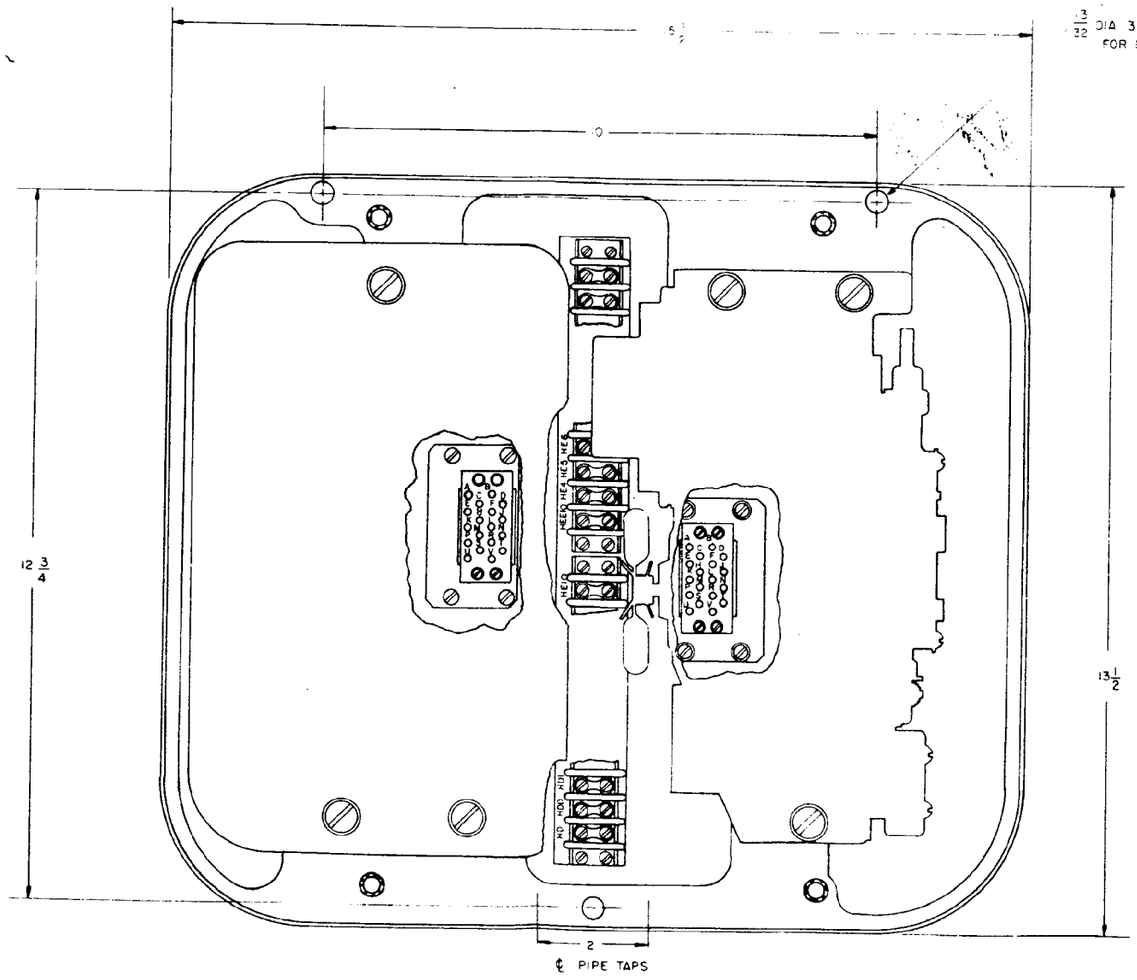
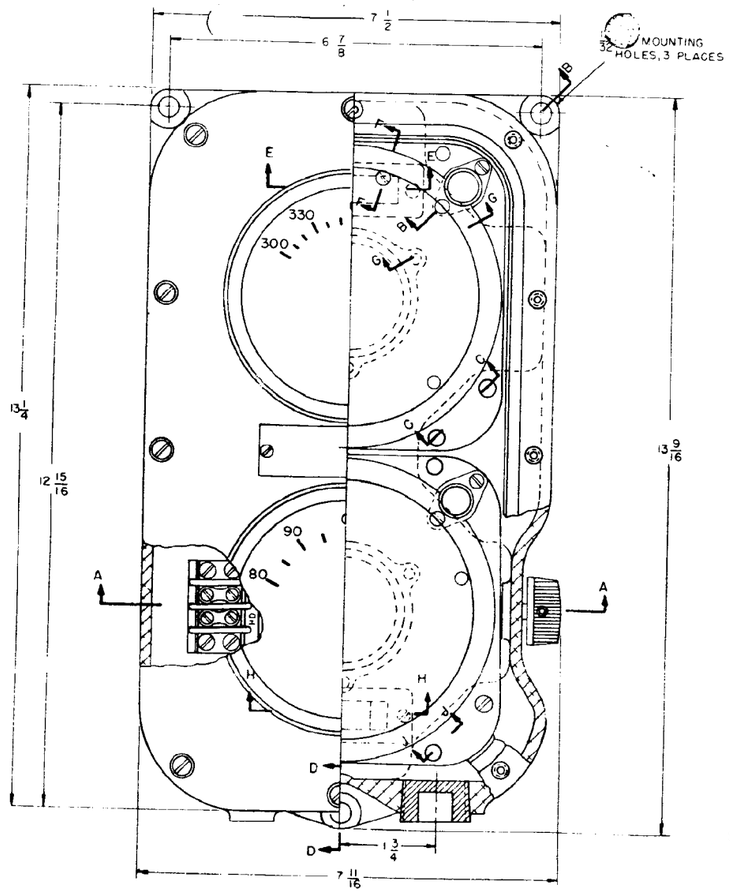
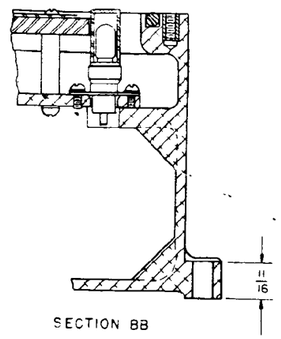


Figure 2- Dimensions for detecting assembly.

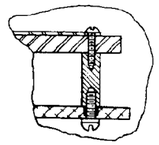
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5 1/2
10
13/32 DIA 3 HOLES FOR MOUNTING
12 3/4
13 1/2
2
℄ PIPE TAPS
HEAD
HEATER
HEATER
HEATER

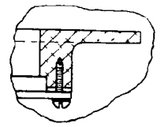




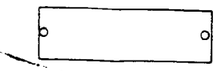
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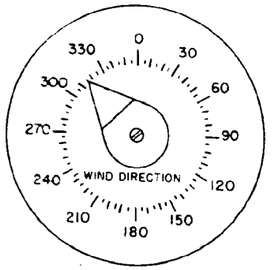
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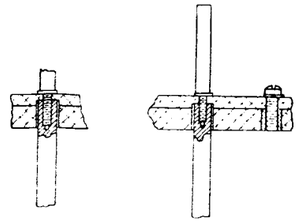
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IDENTIFICATION PLATE

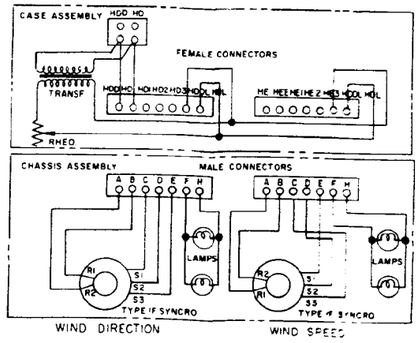


WIND DIRECTION



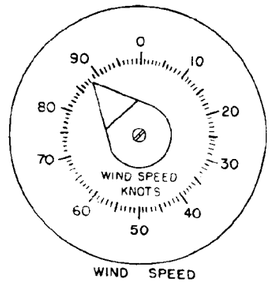
SECTION JJ

SECTION CC

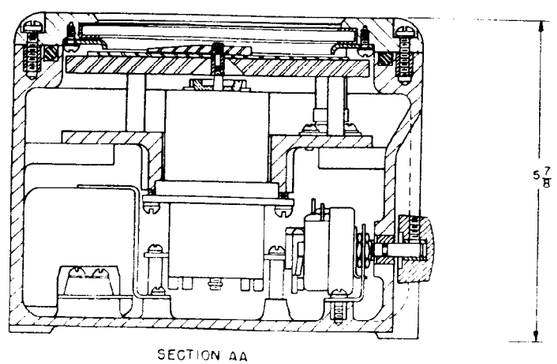


WIRING DIAGRAM

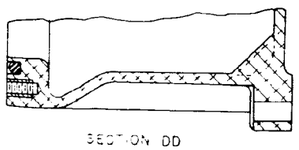
WIND DIRECTION WIND SPEED



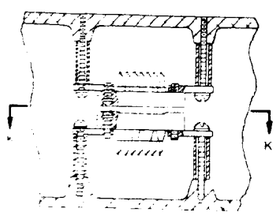
WIND SPEED



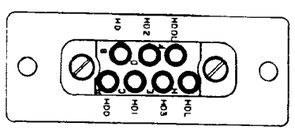
SECTION AA



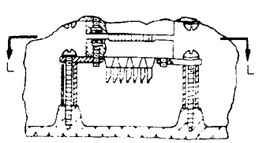
SECTION DD



SECTION EE

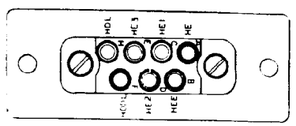


SECTION KK



SECTION HH

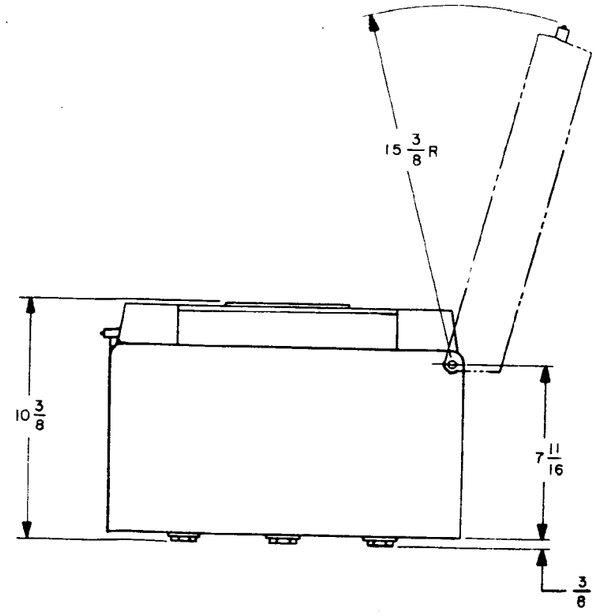
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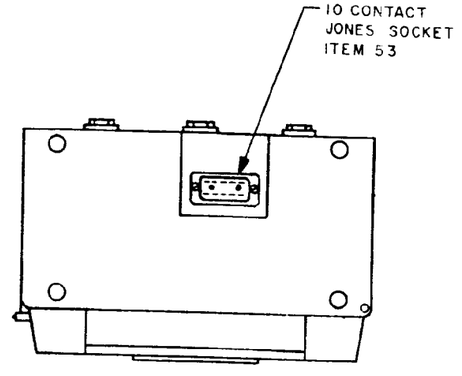


SECTION LL

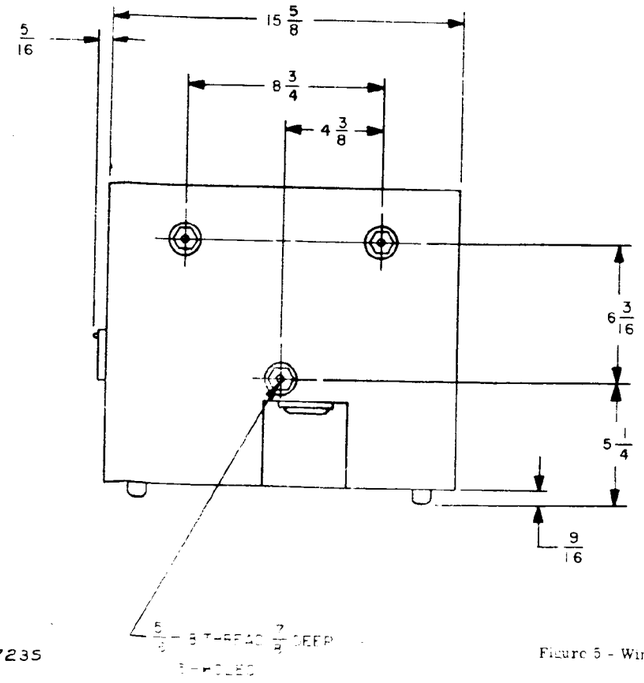
Figure 4 - Outline of wind direction and speed indicator, type B.

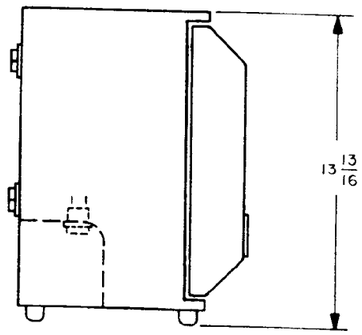
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BOTTOM VIEW





SH 7235

Figure 5 - Wind direction and wind speed recorder, type B (case).

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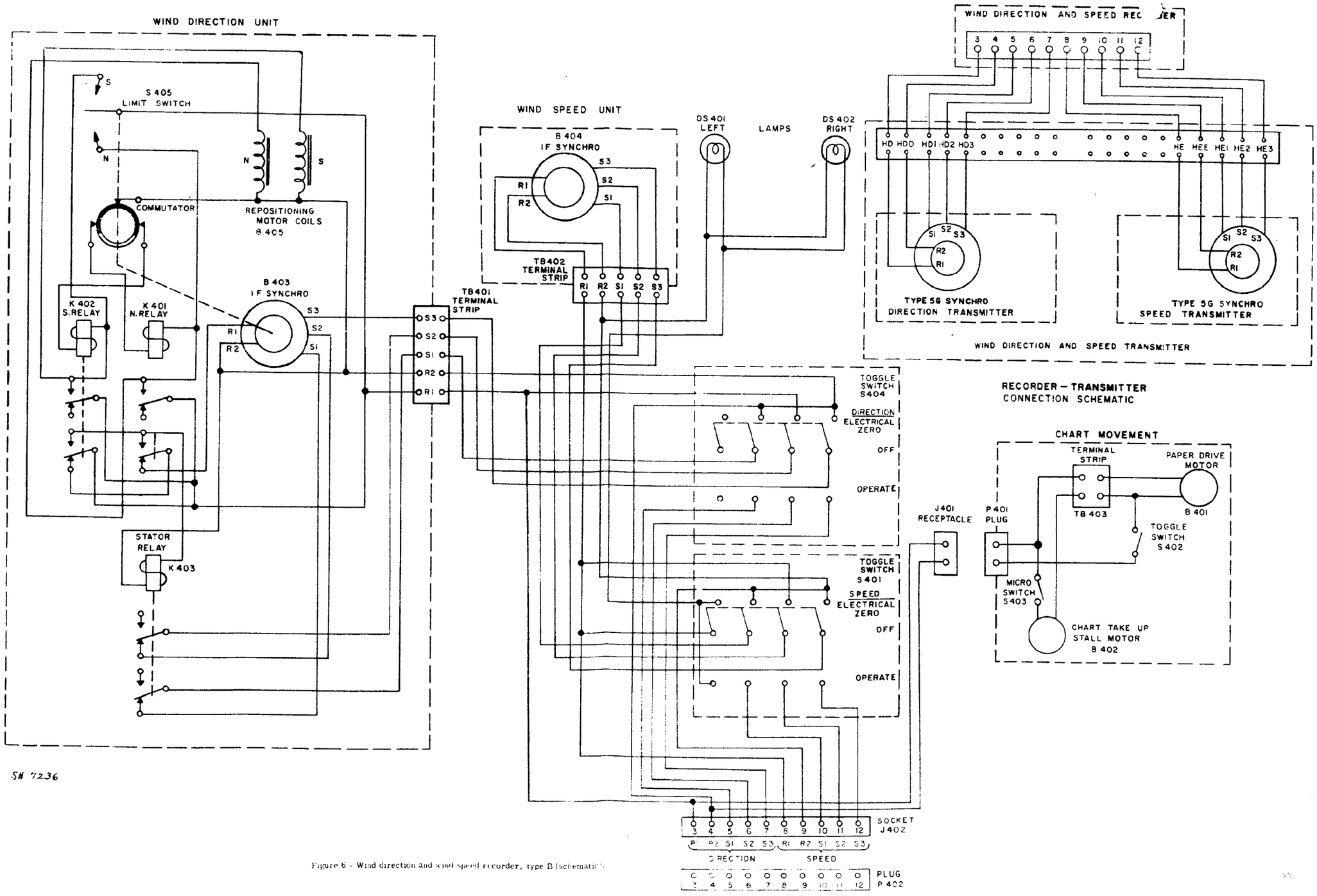
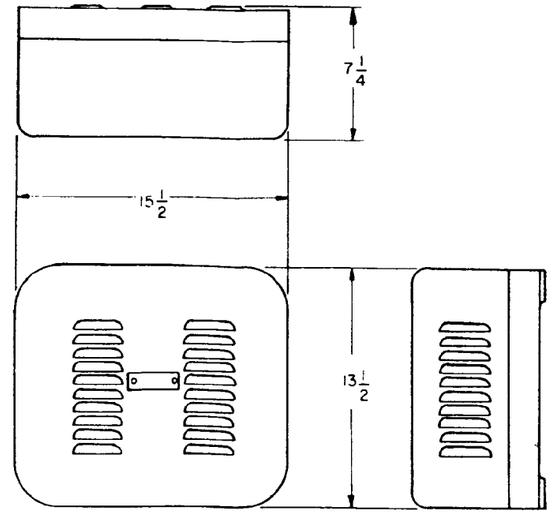


Figure 6 - Wind direction and wind speed recorder, type B (schematic).

RECORDER – TRANSMITTER
CONNECTION SCHEMATIC

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SH 7253

Figure 7 - Computer, Meteorological data, crosswind-headwind.

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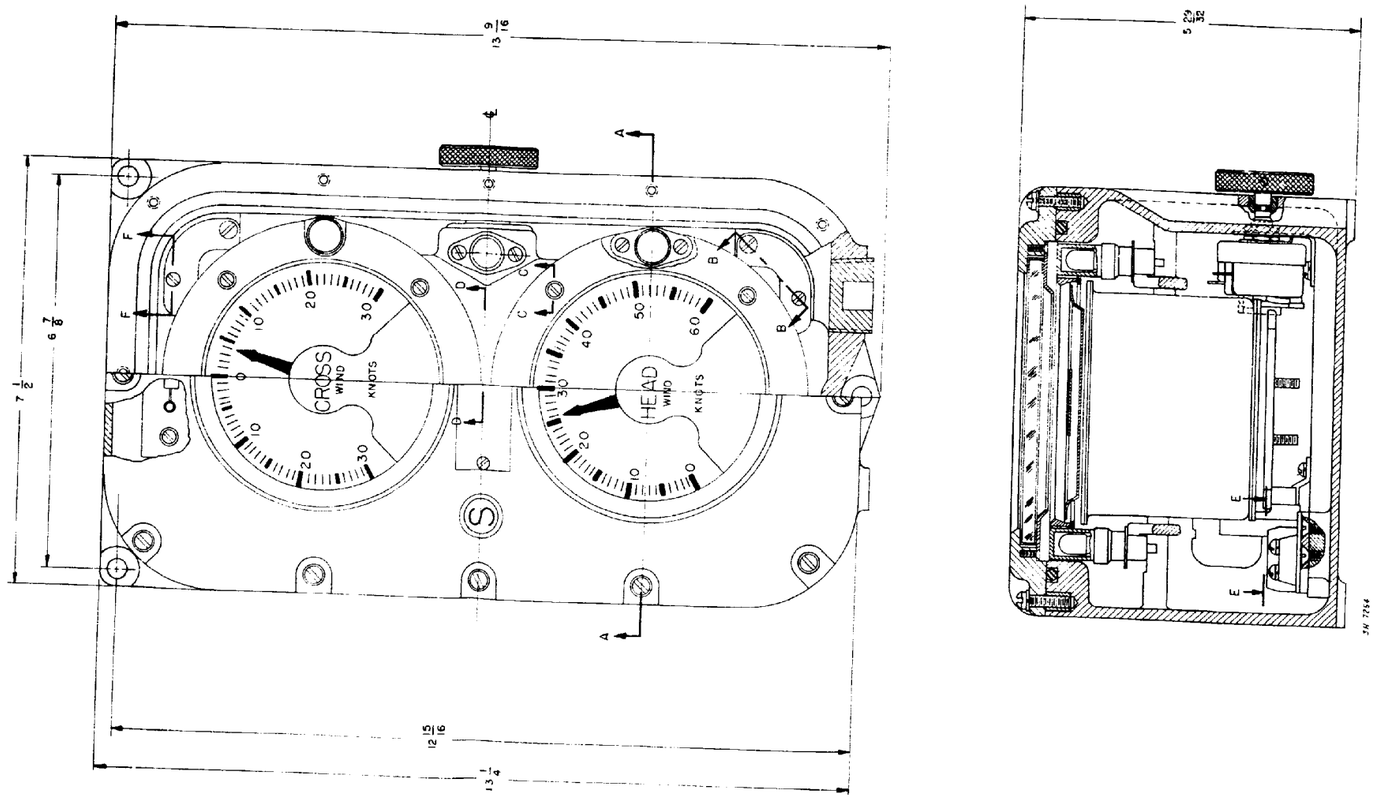


Figure 8. Outline of crosswind-headwind indicator.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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